Chapter 15: Outcome Regression and Propensity Scores

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6 Summary 13

This chapter covers **outcome regression** and **propensity score** methods (stratification, standardization, and matching on the propensity score): the most commonly used parametric methods for causal inference. They work well for treatments fixed at a single time, but they are not designed for the complexities of time-varying treatments, which is why the book presents them only after the g-methods.

This chapter is based on Hernán and Robins (2020, chap. 15, pp. 199-208).

Key theme: outcome regression and propensity score methods need the same identifying conditions as every other method (exchangeability, positivity, and consistency / well-defined interventions), plus correctly specified models for the quantities they estimate along the way. Unlike the g-methods of Chapters 12-14, they have limited applicability to complex longitudinal data (Hernán and Robins 2020, 199).

1 15.1 Outcome Regression (pp. 199-201)

1.1 A Structural Model with Parameters for L

Structural nested models (Chapter 14) contain parameters for A and for $A \times L$ product terms only, so g-estimation is agnostic about the L-Y relation. If instead we are willing to specify the L-Y relation within levels of A, we can use a structural model that also has parameters for L.

Definition 1.1 (Structural Model with Parameters for L)**.** The **structural model with parameters for** L states that, for each $a \in \{0, 1\}$,

$$\mathrm{E}[Y^{a,c=0} \mid L] = \beta_0 + \beta_1 a + a\beta_2{}^\top L + \beta_3{}^\top L,$$

where β_0 and β_1 are scalar parameters and β_2 and β_3 are vector parameters.

In Chapter 12 this model was called a **faux marginal structural model**: it has the form of a marginal structural model, but because it conditions on the entire vector L (not a subset V), the stabilized IP weights $SW^A(L)$ all equal 1 and no weighting is needed (Hernán and Robins 2020, 199).

β_3 is often called the "main effect" of L, but it need not have a causal interpretation (the L-Y relation may itself be confounded); it only describes how the mean of $Y^{a=0,c=0}$ varies with L.

1.2 Conditional Effects from the Structural Model

Proposition 1.1 (Stratum-Specific Effect)**.** *Under the structural model of Definition 1.1, the average causal effect in stratum $L = l$ is $\beta_1 + \beta_2{}^\top l$.*

Proof

Proof. Evaluate the model at $a = 1$ and $a = 0$ and subtract:

$$\begin{aligned}
&\mathrm{E}[Y^{a=1,c=0} \mid L = l] - \mathrm{E}[Y^{a=0,c=0} \mid L = l] \\
&\quad = (\beta_0 + \beta_1 \cdot 1 + 1 \cdot \beta_2{}^\top l + \beta_3{}^\top l) - (\beta_0 + \beta_1 \cdot 0 + 0 \cdot \beta_2{}^\top l + \beta_3{}^\top l) \\
&\quad = (\beta_0 + \beta_1 + \beta_2{}^\top l + \beta_3{}^\top l) - (\beta_0 + \beta_3{}^\top l) \\
&\quad = \beta_1 + \beta_2{}^\top l.
\end{aligned}$$

□

So the stratum-specific effects depend only on (β_1, β_2); the counterfactual means under no treatment depend on (β_0, β_3).

1.3 Estimating the Structural Model by Outcome Regression

Proposition 1.2 (Outcome Regression Identifies the Structural Model)**.** *Suppose that L is discrete (it takes finitely or countably many values), and that, for each $a \in \{0, 1\}$, $\mathrm{E}[|Y^{a,c=0}|] < \infty$. Suppose also that L suffices to adjust for confounding and selection bias, that is, conditional exchangeability $Y^{a,c=0} \perp\!\!\!\perp (A, C) \mid L$ holds for $a \in \{0, 1\}$; that positivity $\Pr[A = a, C = 0 \mid L = l] > 0$ holds for $a \in \{0, 1\}$ and every value l with $\Pr[L = l] > 0$; and that consistency (well-defined interventions) holds: $Y = Y^{a,c=0}$ whenever $A = a$ and $C = 0$.*

1. *The conditional counterfactual mean is identified:*

$$\mathrm{E}[Y^{a,c=0} \mid L = l] = \mathrm{E}[Y \mid A = a, C = 0, L = l],$$

   for $a \in \{0, 1\}$ and every value l with $\Pr[L = l] > 0$.

2. *Suppose further that the structural model of Definition 1.1 holds. Let $X \overset{def}{=}$*

$(1, A, AL^\top, L^\top)^\top$ *be the regressor vector, and stack the structural parameters as* $\beta \overset{def}{=} (\beta_0, \beta_1, {\beta_2}^\top, {\beta_3}^\top)^\top$*, and suppose that* $\mathrm{E}[Y^2 \mid C = 0] < \infty$*, that* $\mathrm{E}\left[\|X\|^2 \mid C = 0\right] < \infty$*, and that the second-moment matrix* $\mathrm{E}[XX^\top \mid C = 0]$ *is invertible. Then the outcome regression model*

$$\mathrm{E}[Y \mid A, C = 0, L] = \alpha_0 + \alpha_1 A + A{\alpha_2}^\top L + {\alpha_3}^\top L$$

with stacked coefficients $\alpha \overset{def}{=} (\alpha_0, \alpha_1, {\alpha_2}^\top, {\alpha_3}^\top)^\top$ *is correctly specified, and its population least-squares coefficients satisfy* $\alpha = \beta$*.*

Proof

Proof. Part 1:

$$\begin{aligned}
\mathrm{E}[Y^{a,c=0} \mid L = l] &= \mathrm{E}[Y^{a,c=0} \mid A = a, C = 0, L = l] && \text{(conditional exchangeability; positivity makes the conditioning e} \\
&= \mathrm{E}[Y \mid A = a, C = 0, L = l] && \text{(consistency).}
\end{aligned}$$

Part 2: by part 1 and the structural model, $\mathrm{E}[Y \mid A = a, C = 0, L = l] = \beta_0 + \beta_1 a + a{\beta_2}^\top l + {\beta_3}^\top l$ for every (a, l) with positive probability among the uncensored, so the linear model with $\alpha = \beta$ equals the conditional mean and is correctly specified; in vector form, $\mathrm{E}[Y \mid A, L, C = 0] = X^\top \beta$. The moment conditions make every expectation in the normal equations finite. The population least-squares coefficients α are the solutions of the normal equations $\mathrm{E}[X(Y - X^\top \alpha) \mid C = 0] = 0$. The vector β solves them:

$$\begin{aligned}
\mathrm{E}[X(Y - X^\top \beta) \mid C = 0] &= \mathrm{E}[\mathrm{E}[X(Y - X^\top \beta) \mid A, L, C = 0] \mid C = 0] && \text{(iterated expectations)} \\
&= \mathrm{E}[X(\mathrm{E}[Y \mid A, L, C = 0] - X^\top \beta) \mid C = 0] && (X \text{ is a function of } (A, L)) \\
&= \mathrm{E}[X(X^\top \beta - X^\top \beta) \mid C = 0] && \text{(correct specification)} \\
&= 0.
\end{aligned}$$

Because $\mathrm{E}[XX^\top \mid C = 0]$ is invertible, the normal equations have a unique solution, so $\alpha = \beta$. □

So the structural parameters can be estimated by ordinary least squares, fitting the outcome regression model of Proposition 1.2.

Like stratification (Chapter 3), outcome regression adjusts for confounding by estimating the effect **within strata of** L. In Section 13.2, the outcome regression was an intermediate step toward a standardized (marginal) mean; here the regression is the end of the procedure: we compare conditional means instead of standardizing them (Hernán and Robins 2020, 200).

Discrete Outcomes

The same approach works for discrete outcomes, e.g., a logistic model for $\Pr[Y = 1 \mid A = a, C = 0, L]$ when Y is dichotomous.

Example 1.1 (Smoking Cessation and Weight Gain (NHEFS))**.** Using the outcome model of Section 13.2, which has a single product term (between A and smoking intensity), the book reports $\hat{\beta}_1 = 2.6$ and $\hat{\beta}_2 = 0.05$ (Hernán and Robins 2020, 200).

- Estimated effect $\hat{\mathrm{E}}[Y \mid A = 1, C = 0, L] - \hat{\mathrm{E}}[Y \mid A = 0, C = 0, L]$: 2.8 kg (95% CI: 1.5, 4.1) for those smoking 5 cigarettes/day; 4.4 kg (95% CI: 2.8, 6.1) for those smoking 40 cigarettes/day.

- Model with **no** product terms (assumes no effect modification by any variable in L): $\widehat{\beta}_1 = 3.5$ kg (95% CI: 2.6, 4.3), which then estimates both the conditional and the marginal effect.

Remark 1.1 (Rounded Coefficients). By Proposition 1.1, each stratum-specific estimate is $\widehat{\beta}_1 + \widehat{\beta}_2 \times$ (cigarettes/day). The book reports the coefficients rounded to one or two decimals, so plugging the rounded values into that formula reproduces the reported effect estimates only approximately (e.g., $2.6 + 0.05 \times 40 = 4.6$ versus the reported 4.4).

Definition 1.2 (Nuisance Parameter)**.** A **nuisance parameter** is a parameter that a model must contain but that is not itself the target of inference.

ℹ Fine Point 15.1: Nuisance Parameters

Two methods for the same effect rely on different nuisance parameters (Definition 1.2):

- **Outcome regression** for β_1, β_2: consistent (in general) only if $\beta_0 + {\beta_3}^\top L$ correctly models how $\mathrm{E}[Y^{a=0,c=0} \mid L]$ depends on L. The nuisance parameters are β_0 and β_3.
- **G-estimation** of the structural nested model $\mathrm{E}[Y^{a,c=0} - Y^{a=0,c=0} \mid L] = \beta_1 a + a{\beta_2}^\top L$: consistent (in general) only if the treatment model for $\Pr[A = 1 \mid L]$ is correct. The nuisance parameters are those of the treatment model, e.g., γ_0, γ_1 in $\operatorname{logit} \Pr[A = 1 \mid L] = \gamma_0 + {\gamma_1}^\top L$.

Example: if L should enter as $\beta_3 L + \beta_4 L^2$ but is modeled as $\beta_3 L$ only, outcome regression is biased; g-estimation is unaffected by that error but is biased if the L-A relation is misspecified. For a time-fixed treatment, choosing between the two amounts to deciding which nuisance model we can specify more accurately. When possible, a better option is a doubly robust method (Fine Point 13.2) (Hernán and Robins 2020, 200).

2 15.2 Propensity Scores (pp. 201-202)

Definition 2.1 (Propensity Score)**.** For a dichotomous treatment A, the **propensity score** is the conditional probability of treatment given the covariates:

$$\pi(L) \stackrel{\text{def}}{=} \Pr[A = 1 \mid L].$$

Example 2.1 (Propensity Score in a Randomized Trial)**.** In an ideal randomized trial assigning half the individuals to $A = 1$, $\pi(L) = 0.5$ for everyone, whatever L is.

Remark 2.1 (The Propensity Score Must Usually Be Estimated). In an observational study, $\pi(L)$ is unknown and must be estimated, e.g., with the same logistic model used for IP weighting and g-estimation in Chapters 12 and 14.

Remark 2.2 (Dichotomous Treatments Only). The book restricts attention to dichotomous treatments: propensity score methods other than IP weighting, g-estimation, and related doubly robust estimators are difficult to generalize to non-dichotomous treatments (Hernán and Robins 2020, 201).

Example 2.2 (Estimated Propensity Scores in NHEFS)**.** From the logistic model for quitting smoking given L (Hernán and Robins 2020, 201):

- lowest estimated propensity score: 0.053 (individual 22941);
- highest: 0.793 (individual 24949);
- mean estimated propensity score: 0.312 among quitters ($A = 1$), 0.245 among nonquitters ($A = 0$).

Figure 15.1 of the book shows the two distributions; they overlap over most of the range.

Quitters having a higher average estimated probability of quitting is exactly what we expect when L predicts quitting.

2.1 The Balancing Property

Definition 2.2 (Balancing Score and Prognostic Score (Technical Point 15.1))**.**

- A **balancing score** $b(L)$ is any function of L such that $A \perp\!\!\!\perp L \mid b(L)$. We then say that $b(L)$ **balances** the covariates L.
- A **prognostic score** $s(L)$ is a function of L such that $Y^{a=0} \perp\!\!\!\perp L \mid s(L)$.

Proposition 2.1 (The Propensity Score Balances L)**.** *Let A be dichotomous and $\pi(L)$ its propensity score (Definition 2.1). Then $\pi(L)$ is a balancing score (Definition 2.2):*

$$A \perp\!\!\!\perp L \mid \pi(L).$$

Proof

Proof. The book argues graphically (Technical Point 15.1); the same result follows from iterated expectations. Since $\pi(L)$ is a function of L, conditioning on $(L, \pi(L))$ is the same as conditioning on L:

$$\Pr[A = 1 \mid L, \pi(L)] = \Pr[A = 1 \mid L] = \pi(L).$$

Conditioning on $\pi(L)$ alone, by the law of iterated expectations:

$$\begin{aligned} \Pr[A = 1 \mid \pi(L)] &= \mathrm{E}[\Pr[A = 1 \mid L] \mid \pi(L)] \\ &= \mathrm{E}[\pi(L) \mid \pi(L)] \\ &= \pi(L). \end{aligned}$$

The two probabilities are equal, so A does not depend on L once $\pi(L)$ is fixed: $A \perp\!\!\!\perp L \mid \pi(L)$. □

In words: individuals with the same $\pi(L)$ may differ in L (e.g., smoking intensity and exercise), but among all individuals with a given value of $\pi(L)$ in the super-population, the distribution of L is the same in the treated and the untreated. A function $b(L)$ is a balancing score exactly when $\pi(L)$ can be written as a function of $b(L)$ (Rosenbaum and Rubin 1983, Theorem 2), so the propensity score is the coarsest balancing score.

If the distribution of $\pi(L)$ were the same in the treated and the untreated, there would be no association between L and A: then $A \perp\!\!\!\perp \pi(L)$, and combining this with $A \perp\!\!\!\perp L \mid \pi(L)$ from Proposition 2.1 gives $A \perp\!\!\!\perp L$.

Limits of Propensity Score Balance

Two caveats from the book (Hernán and Robins 2020, 201–2):

- The propensity score balances only the **measured** covariates L; unlike randomization, it does nothing about unmeasured confounders.
- In a finite study population the **true** propensity score balances L only approximately, because of sampling variability. An estimated propensity score from a correct model generally gives better balance; in a randomized experiment, adjusting for the estimated $\pi(L)$ corrects both systematic and chance imbalances, whereas the true $\pi(L)$ ignores the chance imbalances.

Rosenbaum and Rubin proved that exchangeability and positivity given L imply exchangeability and positivity given any balancing score (Rosenbaum and Rubin 1983, Theorem 3; Hernán and Robins 2020, 202). Graphically, $\pi(L)$ sits between L and A, with a deterministic arrow $L \to \pi(L)$ (book Figure 15.2); conditioning on $\pi(L)$ blocks the backdoor paths from A to Y through L ($A \leftarrow \pi(L) \leftarrow L \to Y$), so adjusting for it suffices.

Methods based on prognostic scores need stronger assumptions and do not extend readily to time-varying treatments; see Hansen (2008) and Abadie et al. (2013), as cited in Hernán and Robins (2020, 202).

Theorem 2.1 (Exchangeability and Positivity Given the Propensity Score)**.** *Let A be dichotomous and $\pi(L)$ its propensity score (Definition 2.1).*

1. *Fix $a \in \{0,1\}$. If $Y^a \perp\!\!\!\perp A \mid L$, then $Y^a \perp\!\!\!\perp A \mid \pi(L)$.*
2. *Suppose L is discrete. Then positivity holds within levels of $\pi(L)$, that is, $\Pr[A = a \mid \pi(L) = s] > 0$ for both $a \in \{0,1\}$ and every s with $\Pr[\pi(L) = s] > 0$, **if and only if** positivity holds within levels of L, that is, $\Pr[A = a \mid L = l] > 0$ for both $a \in \{0,1\}$ and every l with $\Pr[L = l] > 0$.*

Proof (Proof of part 1)

Proof of part 1.

$$
\begin{aligned}
\Pr[A = 1 \mid Y^a, \pi(L)] &= \mathrm{E}[\Pr[A = 1 \mid Y^a, L] \mid Y^a, \pi(L)] && \text{(iterated expectations; } \pi(L) \text{ is a function of } L) \\
&= \mathrm{E}[\Pr[A = 1 \mid L] \mid Y^a, \pi(L)] && (Y^a \perp\!\!\!\perp A \mid L) \\
&= \mathrm{E}[\pi(L) \mid Y^a, \pi(L)] && \text{(definition of } \pi(L)) \\
&= \pi(L),
\end{aligned}
$$

which does not depend on Y^a; hence $Y^a \perp\!\!\!\perp A \mid \pi(L)$. □

Proof (Proof of part 2)

Proof of part 2. The proof of Proposition 2.1 showed $\Pr[A = 1 \mid \pi(L)] = \pi(L)$, so $\Pr[A = 1 \mid \pi(L) = s] = s$ and $\Pr[A = 0 \mid \pi(L) = s] = 1 - s$ for every s with $\Pr[\pi(L) = s] > 0$. Because L is discrete, $\Pr[\pi(L) = s] = \sum_{l:\pi(l)=s} \Pr[L = l]$, so the values s with $\Pr[\pi(L) = s] > 0$ are exactly the values $\pi(l)$ with $\Pr[L = l] > 0$. Positivity within levels of $\pi(L)$ therefore holds if and only if $0 < \pi(l) < 1$ for every l with $\Pr[L = l] > 0$. Positivity within levels of L requires $\Pr[A = 1 \mid L = l] = \pi(l)$ and $\Pr[A = 0 \mid L = l] = 1 - \pi(l)$ to be positive for every l with $\Pr[L = l] > 0$, which is the same condition. □

Hence, if L is sufficient to adjust for confounding, so is $\pi(L)$ (Hernán and Robins 2020, 202); part 2 of Theorem 2.1 states the positivity condition given $\pi(L)$.

2.2 Using the Propensity Score

Algorithm 2.1 (Adjusting for the Propensity Score)**.** *Suppose exchangeability and positivity hold given $\pi(L)$, consistency holds, and the model for $\Pr[A = 1 \mid L]$ is correctly specified; for outcome regression on, or standardization over, $\hat{\pi}(L)$, suppose also that the model for the relation between $\pi(L)$ and Y is correctly specified. Then causal effects can be estimated as follows.*

1. *Fit a model for $\Pr[A = 1 \mid L]$.*
2. *Compute the estimated propensity score $\hat{\pi}(L_i)$ for each individual i.*
3. *Use $\hat{\pi}(L)$ in place of L: stratify on $\hat{\pi}(L)$ (including outcome regression on it), standardize over $\hat{\pi}(L)$, or match on $\hat{\pi}(L)$.*

3 15.3 Propensity Stratification and Standardization (pp. 202-204)

Proposition 3.1 (Effects Within Levels of the Propensity Score)**.** *Let A be dichotomous and $\pi(L)$ its propensity score. Suppose that L is discrete (so $\pi(L)$ is discrete too), and that, for each $a \in \{0, 1\}$, $\mathrm{E}[|Y^{a,c=0}|] < \infty$. Suppose also conditional exchangeability given the propensity score, $Y^{a,c=0} \perp\!\!\!\perp (A, C) \mid \pi(L)$ for $a \in \{0, 1\}$; positivity, $\Pr[A = a, C = 0 \mid \pi(L) = s] > 0$ for $a \in \{0, 1\}$ and every s with $\Pr[\pi(L) = s] > 0$; and consistency, $Y = Y^{a,c=0}$ whenever $A = a$ and $C = 0$. Then, for each value s of the propensity score with $\Pr[\pi(L) = s] > 0$,*

$$\mathrm{E}[Y^{a=1,c=0} \mid \pi(L) = s] - \mathrm{E}[Y^{a=0,c=0} \mid \pi(L) = s] = \mathrm{E}[Y \mid A = 1, C = 0, \pi(L) = s] - \mathrm{E}[Y \mid A = 0, C = 0, \pi(L) = s].$$

Proof

Proof. For each $a \in \{0, 1\}$,

$$\begin{aligned} \mathrm{E}[Y^{a,c=0} \mid \pi(L) = s] &= \mathrm{E}[Y^{a,c=0} \mid A = a, C = 0, \pi(L) = s] && \text{(exchangeability given } \pi(L)\text{; positivity)} \\ &= \mathrm{E}[Y \mid A = a, C = 0, \pi(L) = s] && \text{(consistency).} \end{aligned}$$

Subtracting the $a = 0$ line from the $a = 1$ line gives the result. □

For treatment alone, without censoring, Theorem 2.1 shows that exchangeability and positivity given L carry over to $\pi(L)$. When no one is censored, the joint (A, C) conditions of Proposition 3.1 reduce to these treatment-only conditions, so they hold whenever exchangeability and positivity given L hold (with L discrete, as in Proposition 3.1). With censoring, $\pi(L)$ alone need not suffice; one would instead condition on a score $b(L)$ with $(A, C) \perp\!\!\!\perp L \mid b(L)$, for example the vector of joint probabilities $\Pr[A = a, C = c \mid L]$ over all (a, c).

Few Individuals Share a Propensity Score Value

In practice $\pi(L)$ is generally continuous, so almost no two individuals share a value s (e.g., only one NHEFS individual had an estimated $\pi(L)$ of 0.6563).

3.1 Stratifying on Deciles

Example 3.1 (Decile Stratification in NHEFS)**.** Classify individuals into 10 strata (deciles) of the estimated $\pi(L)$, about 162 individuals each, and estimate the effect within each decile (Hernán and Robins 2020, 203):

- decile-specific effects of smoking cessation on weight gain ranged from 0.0 to 6.6 kg, with wide 95% CIs;
- the same estimates come from an outcome regression of Y on A, 9 decile indicators, and 9 $A\times$ indicator product terms;
- without product terms (no effect modification by $\pi(L)$, as most applications assume): 3.5 kg (95% CI: 2.6, 4.4).

With wide confidence intervals, differences between deciles may reflect chance.

3.2 A Problem with Categorized Propensity Scores

⚠ Residual Imbalance Within Propensity Strata

Within a decile, the distribution of the continuous $\pi(L)$ may still differ between treated and untreated (e.g., higher average $\pi(L)$ in the treated), so the groups may not be exchangeable within that decile.

💡 Use the Propensity Score as a Continuous Covariate

Remedy: use the estimated $\pi(L)$ as a **continuous** covariate in the outcome model $\mathrm{E}[Y \mid A, C = 0, \pi(L)]$.

Example 3.2 (Continuous Propensity Score in NHEFS)**.** In NHEFS (linear in $\pi(L)$): 3.6 kg (95% CI: 2.7, 4.5).

IP weighting and g-estimation never had this problem, because they used the numerical value of the estimated probability rather than a categorized version (Hernán and Robins 2020, 203).

⚠ The IP Weight Denominator Is a Function of the Propensity Score

For a dichotomous treatment the IP weight denominator is not $\pi(L)$ itself but a function of it: $\pi(L)$ for the treated ($A = 1$) and $1 - \pi(L)$ for the untreated ($A = 0$).

The outcome model now requires a correct specification of the $\pi(L)$-Y relation. Because $\pi(L)$ is one-dimensional, this is easy to protect with flexible terms such as cubic splines; IP weighting and g-estimation were agnostic about that relation. Note, though, that $\pi(L)$ itself must still be estimated from a model of A on the high-dimensional L, as in IP weighting and g-estimation.

3.3 Standardizing over the Propensity Score

The outcome model on $\pi(L)$ estimates effects **within** levels s of $\pi(L)$.

Algorithm 3.1 (Standardization over the Propensity Score)**.** *For the population average causal effect* $\mathrm{E}[Y^{a=1,c=0}] - \mathrm{E}[Y^{a=0,c=0}]$*, standardize the fitted conditional means over the distribution of an estimated adjustment score* $b(L)$*: the Chapter 13 procedure with* $b(L)$ *in place of* L*. When no one is censored,* $b(L) = \pi(L)$*, which suffices for adjustment when*

exchangeability and positivity hold given L (Theorem 2.1). With censoring, $b(L)$ is a joint score with $(A, C) \perp\!\!\!\perp L \mid b(L)$, as discussed after Proposition 3.1, such as the vector of joint probabilities $\Pr[A = a, C = c \mid L]$ over all (a, c). This joint score suffices when, for each $a \in \{0, 1\}$, $Y^{a,c=0} \perp\!\!\!\perp (A, C) \mid L$ and $\Pr[A = a, C = 0 \mid L] > 0$: because $(A, C) \perp\!\!\!\perp L \mid b(L)$, these conditions carry over from L to $b(L)$, by the argument Theorem 2.1 gives for A alone. In either case, consistency must hold, and the treatment, censoring (if any), and outcome models must be correctly specified.

1. *Estimate the score $\hat{b}(L_i)$ for each individual $i = 1, \ldots, n$, censored or not. Without censoring, $\hat{b}(L_i) = \hat{\pi}(L_i)$. With censoring, combine a treatment model and a censoring model, $\widehat{\Pr}[A = a, C = c \mid L] = \widehat{\Pr}[A = a \mid L]\,\widehat{\Pr}[C = c \mid L, A = a]$ for each (a, c).*
2. *Fit an outcome model for $\mathrm{E}[Y \mid A, C = 0, b(L)]$ once, among the uncensored, with $\hat{b}(L)$ as the covariate(s); write $\hat{m}(a, s)$ for its fitted value at $A = a$ and score value s.*
3. *For each $a \in \{0, 1\}$, predict each individual's outcome at $A = a$ and their own $\hat{b}(L_i)$, and average the predictions over all n individuals:*

$$\hat{\mathrm{E}}[Y^{a,c=0}] = \frac{1}{n} \sum_{i=1}^{n} \hat{m}(a, \hat{b}(L_i)).$$

4. *Estimate the average causal effect by the difference $\hat{\mathrm{E}}[Y^{a=1,c=0}] - \hat{\mathrm{E}}[Y^{a=0,c=0}]$.*

Example 3.3 (Standardized Estimate in NHEFS)**.** Standardizing over the estimated propensity score (the outcome model may include an $A \times \pi(L)$ product term) gives 3.6 kg (95% CI: 2.7, 4.6) (Hernán and Robins 2020, 203).

4 15.4 Propensity Matching (pp. 204-205)

Matching on $\pi(L)$ is analogous to matching on a single continuous variable (Chapter 4).

Definition 4.1 (Propensity-Matched Population)**.** A **propensity-matched population** is a subset of the study population, possibly reweighted, in which the distribution of $\pi(L)$ is the same among the treated and the untreated, or approximately the same when matches need only fall within a caliper.

- E.g., pair each treated individual with one (or more) untreated individuals with the same, or a close, propensity score.
- Under exchangeability and positivity given $\pi(L)$, exact matching on the true $\pi(L)$ makes the treated and the untreated in the matched population exchangeable, so, under consistency, association measures in the matched population equal the effect measures **in the matched population**.
- In practice we match on an estimate $\hat{\pi}(L)$. Equal estimated scores need not mean equal true scores, so even with a correctly specified propensity model this exchangeability holds only approximately, with the approximation improving as the sample grows.
- Caliper matching adds a second approximation: some bias may remain, and it typically shrinks as the caliper narrows.
- With censoring, outcomes are observed only for the uncensored members of the matched population. One way to account for that is to weight the uncensored by the inverse of $\Pr[C = 0 \mid L, A]$ (Chapter 12), which needs a correct censoring model, $Y^{a,c=0} \perp\!\!\!\perp C \mid A, L$, and $\Pr[C = 0 \mid L, A] > 0$.

The matched population can be given the $\pi(L)$ distribution of the treated, of the untreated, or any other distribution (Hernán and Robins 2020, 204).

Variance estimation for matched estimators used to be an open problem; Abadie and Imbens (2006) solved it (Hernán and Robins 2020, 204).

Fine Point 15.2: Effect Modification and the Propensity Score

- Matched and unmatched estimates may differ because of **effect modification**: when untreated far outnumber treated, matching keeps nearly all treated and drops many untreated, so the matched population resembles the treated and the estimate approaches the **effect in the treated** (Technical Point 4.1 gives IP weighting and standardization alternatives for that effect).
- Effect modification across propensity strata may suggest that decision makers treat those likely to benefit (Kurth et al. 2006, as cited in Hernán and Robins (2020, 206)), but statements like "beneficial for propensity scores between 0.11 and 0.93" have little policy relevance because they are not expressed in terms of measured variables L.
- Other reasons matched and overall estimates may differ: positivity violations among the unmatched, or an unmeasured confounder that is more (or less) prevalent, or better (or worse) measured, in the matched population; apparent effect modification may reflect different residual confounding across propensity strata.

4.1 Defining Closeness

Exact matches on a continuous $\pi(L)$ are rare, so we match on a **close** value, e.g., untreated individuals within ± 0.05 of the treated individual's estimated $\pi(L)$ (e.g., a treated individual with estimated $\pi(L) = 0.6563$ could be matched to an untreated individual with 0.6579).

Choosing How Close a Match Must Be

Bias-variance trade-off:

- criteria too loose: the $\pi(L)$ distribution differs between matched treated and untreated, so exchangeability fails;
- criteria too tight: many individuals are excluded; approximate exchangeability, but wider confidence intervals.

4.2 Matching and Positivity

Example 4.1 (Matching and Positivity in NHEFS)**.**

- In NHEFS, the estimated $\pi(L)$ distributions overlapped over most of the range; only 2 treated individuals had values above those of every untreated individual.
- Outcome regression on $\pi(L)$ (Section 15.3) kept them, implicitly treating the lack of comparable untreated individuals as **random nonpositivity**.
- Most matched analyses would drop them.

Matching Ignores the Type of Nonpositivity

Matching does not distinguish random from structural nonpositivity.

The book also states this count as a percentage of the study population (0.01%); that percentage is not reproduced here because it is inconsistent with the counts: 2 individuals are about 0.1% of a study population of roughly 1,600.

4.3 Who Is in the Matched Population?

Remark 4.1 (Target Population of a Matched Analysis).

- In theory, matching targets a well-characterized population: matching every treated individual and dropping unmatched untreated estimates the **effect in the treated** (Fine Point 15.2).
- In practice, treated individuals without a close match are excluded too, so the estimate refers to a subset defined by the values of the estimated $\pi(L)$ that found matches.

⚠ Propensity-Defined Populations Are Hard to Describe

Restricting to, say, estimated $\pi(L) < 0.67$ yields a population that is hard to describe: "individuals do not come with a propensity score tattooed on their forehead" (Hernán and Robins 2020, 205), so transportability to other populations is hard to judge.

4.4 Restrict on Real-World Variables Instead

Example 4.2 (Restricting on Age and Smoking Duration in NHEFS)**.** In NHEFS, the 2 treated individuals with estimated $\pi(L) > 0.67$ were the only participants over age 50 who had smoked for fewer than 10 years. Excluding them gives an estimate for

- smokers under 50, and
- smokers 50 and over who had smoked for at least 10 years,

which is a far more natural target population than "estimated $\pi(L) < 0.67$" (Hernán and Robins 2020, 205).

⚠ Restricting to the Overlap Region

Using propensity scores to **find** the region of overlap can be useful, but restricting the study population to that region is a "lazy" way to secure positivity: the automatic positivity of matching must be weighed against the difficulty of assessing transportability (Hernán and Robins 2020, 205). Even if everyone's propensity score were visible, the population could remain ill-characterized, because the same propensity score value can mean different things in different settings.

5 15.5 Propensity Models, Structural Models, Predictive Models (pp. 205-208)

Part II used two kinds of models for causal inference.

Definition 5.1 (Propensity Models and Structural Models)**.**

- **Propensity models**: models for $\Pr[A \mid L]$, where L are the variables needed for conditional exchangeability.
- **Structural models**: models relating A to some feature (e.g., the mean) of the counterfactual Y^a, marginally or within levels of L. For continuous treatments they are often called **dose-response models**.

Remark 5.1 (Which Parameters Have a Causal Interpretation). We fit a propensity model only to compute the score, so its coefficients are nuisance parameters (Definition 1.2): they describe the association between L and A, not the effect of A. The treatment parameters of a

structural model **do** have a causal interpretation.

Propensity models have been used for matching and stratification (this chapter), IP weighting (Chapter 12), and g-estimation (Chapter 14). Their parameters lack a causal interpretation because L and A can be associated for many reasons: the association reflects the effect of L on A under the book's Figure 7.1, but not under Figures 7.2 and 7.3 (Hernán and Robins 2020, 205).

5.1 Two Classes of Structural Models

Remark 5.2 (Marginal and Nested Structural Models).

- **Marginal structural models**: parameters for treatment, for chosen effect modifiers V, and for $A \times V$ product terms. V reflects only the investigator's interest in effect modification (Section 12.5).
  - No V: truly marginal.
  - $V = L$ (all covariates): a **faux** marginal structural model, which is what outcome regression (Section 15.1) estimates.
- **Structural nested models**: parameters for treatment and for $A \times L$ product terms for those variables in L that are effect modifiers.

See Fine Point 14.1 of the book for the relation between structural nested models and faux semiparametric marginal structural models (Hernán and Robins 2020, 206).

5.2 Causal Versus Predictive Uses of Outcome Regression

Remark 5.3 (Outcome Regression for Prediction). Outcome regression is also widely used for **prediction**:

- online retailers predicting who will buy;
- clinical algorithms identifying patients at high risk.

There, no variable is "treatment" and the parameters need not have any causal interpretation.

Example 5.1 (Prediction Without Causation)**.** Example from the book's margin: Facebook Likes predict sexual orientation, political views, and personality traits (Kosinski et al. 2013, as cited in Hernán and Robins (2020, 206)); that is prediction, not necessarily causation.

5.3 Variable Selection: Prediction Tools Misapplied

⚠ Prediction Tools Can Harm Causal Models

Forward selection, backward elimination, stepwise selection, and machine learning are built to **improve prediction**. Applied to propensity or structural models, they can add superfluous or harmful covariates, causing

- inflated variances, and
- greater bias.

A Propensity Model Need Not Predict Treatment Well

It is tempting to judge a propensity model by how accurately it predicts A. It need not; it only needs to include the L that guarantee exchangeability.

Reporting predictive measures such as Mallows's C_p for propensity models is common, but their relevance for causal inference is questionable (Hernán and Robins 2020, 207). Covariates strongly associated with A but not needed for exchangeability do not reduce bias; including them can yield very large variances or even amplify existing bias.

Example 5.2 (Hospitals Aceso and Panacea)**.** Everyone attends hospital Aceso, where 99% receive $A = 1$, or hospital Panacea, where 99% receive $A = 0$. Hospital affects the outcome only through A, so it is not needed for exchangeability (Hernán and Robins 2020, 207).

- Adding Hospital to the propensity model gives $\pi(L) \approx 0.99$ in Aceso and ≈ 0.01 in Panacea in the target population.
- By chance, some L-strata of the study population may then contain only treated (Aceso) or only untreated (Panacea) individuals.
- Result: a near-infinite variance with **no** reduction in confounding. That A is now well predicted is irrelevant.

In the limit of perfect prediction, every treated individual has $\pi(L) = 1$ and every untreated individual has $\pi(L) = 0$: there is no overlap and the analysis is impossible (Hernán and Robins 2020, 207).

5.4 Self-Inflicted Bias and Model Specification

⚠ Good Predictors Can Be Bad Adjustment Variables

- Including **colliders** can cause systematic bias, even though colliders may be good predictors.
- Including **instruments** (Chapter 16) can amplify bias from unmeasured confounders.
- Chapter 18 returns to these issues.

Remark 5.4 (Requirements Shared by Propensity and Structural Models). Both propensity and structural models require a correctly specified functional form for the covariates (flexible terms such as cubic splines help), plus exchangeability, positivity, and well-defined interventions. Chapter 16 presents a method that does not require exchangeability of treatment.

6 Summary

Table 1: Methods of this chapter and Chapter 14, the models they fit, and what each requires beyond the identifying conditions. All of these methods also need exchangeability, positivity, and consistency. Outcome regression handles censoring by fitting its model among the uncensored; the other methods need a model for censoring as well, except that g-estimation can skip the censoring weights under the conditions noted in Chapter 14.

Method	Model used	Additional modeling requirement
Outcome regression (faux MSM)	$\mathrm{E}[Y \mid A, C = 0, L]$	Correct outcome model (nuisance parameters β_0, β_3; Definition 1.2)
G-estimation (SNM)	$\Pr[A = 1 \mid L]$; with censoring, also $\Pr[C = 0 \mid L, A]$ for the censoring weights	Correct treatment model and correct structural nested model; with censoring, also a correct censoring model

Method	Model used	Additional modeling requirement
Propensity stratification / outcome regression on $\pi(L)$	$\pi(L)$ and $\mathrm{E}[Y \mid A, C=0, \pi(L)]$; with censoring, the joint score $b(L)$ in place of $\pi(L)$	Correct propensity model and correct score-Y relation; with censoring, also a correct censoring model
Propensity standardization	as above	as above; targets the population effect
Propensity matching	$\pi(L)$; with censoring, also $\Pr[C=0 \mid L, A]$	Correct propensity model; with censoring, also a correct censoring model; estimate refers to the matched population

Example 6.1 (NHEFS Estimates Compared)**.** NHEFS estimates of the effect of smoking cessation on weight gain (kg):

- outcome regression, no product terms: 3.5 (2.6, 4.3);
- propensity deciles, no product terms: 3.5 (2.6, 4.4);
- continuous propensity score: 3.6 (2.7, 4.5);
- standardized over the propensity score: 3.6 (2.7, 4.6).

Takeaways (Hernán and Robins 2020, 199–208):

- Outcome regression and propensity score methods need the same identifying conditions as the g-methods, plus correct models for their nuisance parameters (Definition 1.2).
- The propensity score balances measured covariates only.
- Matching guarantees positivity automatically but can produce a target population that is hard to characterize; restricting on real-world variables is preferable.
- Propensity models should include the variables needed for exchangeability, not the best predictors of treatment.
- Unlike the g-methods, these methods do not extend readily to time-varying treatments (Part III).

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://miguelhernan.org/whatifbook.

Rosenbaum, Paul R, and Donald B Rubin. 1983. "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika* 70 (1): 41–55. https://doi.org/10.1093/biomet/70.1.41.